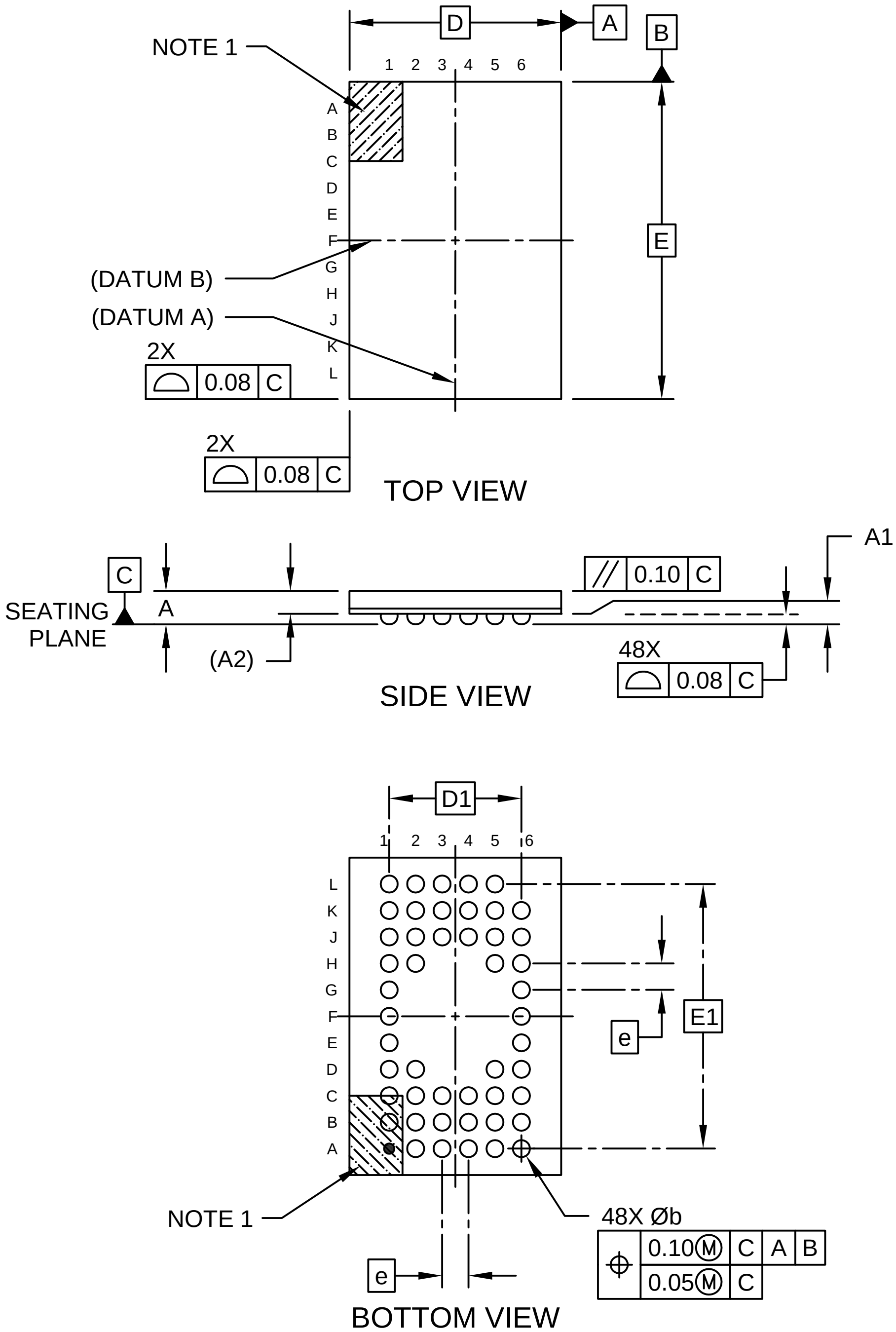
48-Ball Very, Very Thin Fine Pitch Ball Grid Array (CM) - 4x6 mm Body [WFBGA]
SST LEGACY M1QE and M1QF

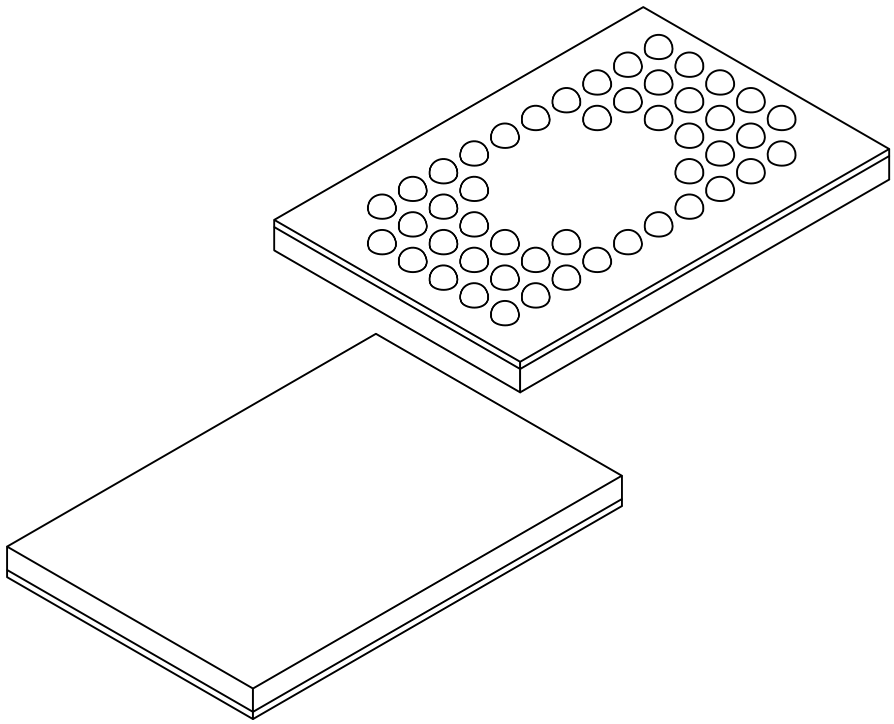
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-165-CM Rev B Sheet 1 of 2

48-Ball Very, Very Thin Fine Pitch Ball Grid Array (CM) - 4x6 mm Body [WFBGA]
SST LEGACY M1QE and M1QF

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



		Units	MILLIMETERS		
		Dimension Limits	MIN	NOM	MAX
Number of Terminals		N		48	
Pitch		e		0.50 BSC	
Overall Height		A	0.53	0.63	0.73
Standoff		A1	0.14	0.20	0.26
Package Thickness		A2		0.43 REF	
Overall Length		D		4.00 BSC	
Overall Terminal Spacing		D1		2.50 BSC	
Overall Width		E		6.00 BSC	
Overall Terminal Spacing		E1		5.00 BSC	
Terminal Width		b	0.27	0.32	0.37

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M

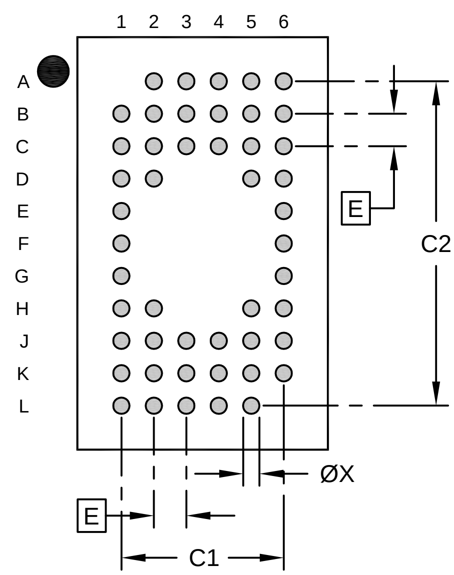
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-165-CM Rev B Sheet 2 of 2

Land Pattern

48-Ball Very, Very Thin Fine Pitch Ball Grid Array (CM) - 4x6 mm Body [WFBGA] SST LEGACY M1QE and M1QF

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		2.50	
Contact Pad Spacing	C2		5.00	
Contact Pad Diameter (X48)	X			0.25

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

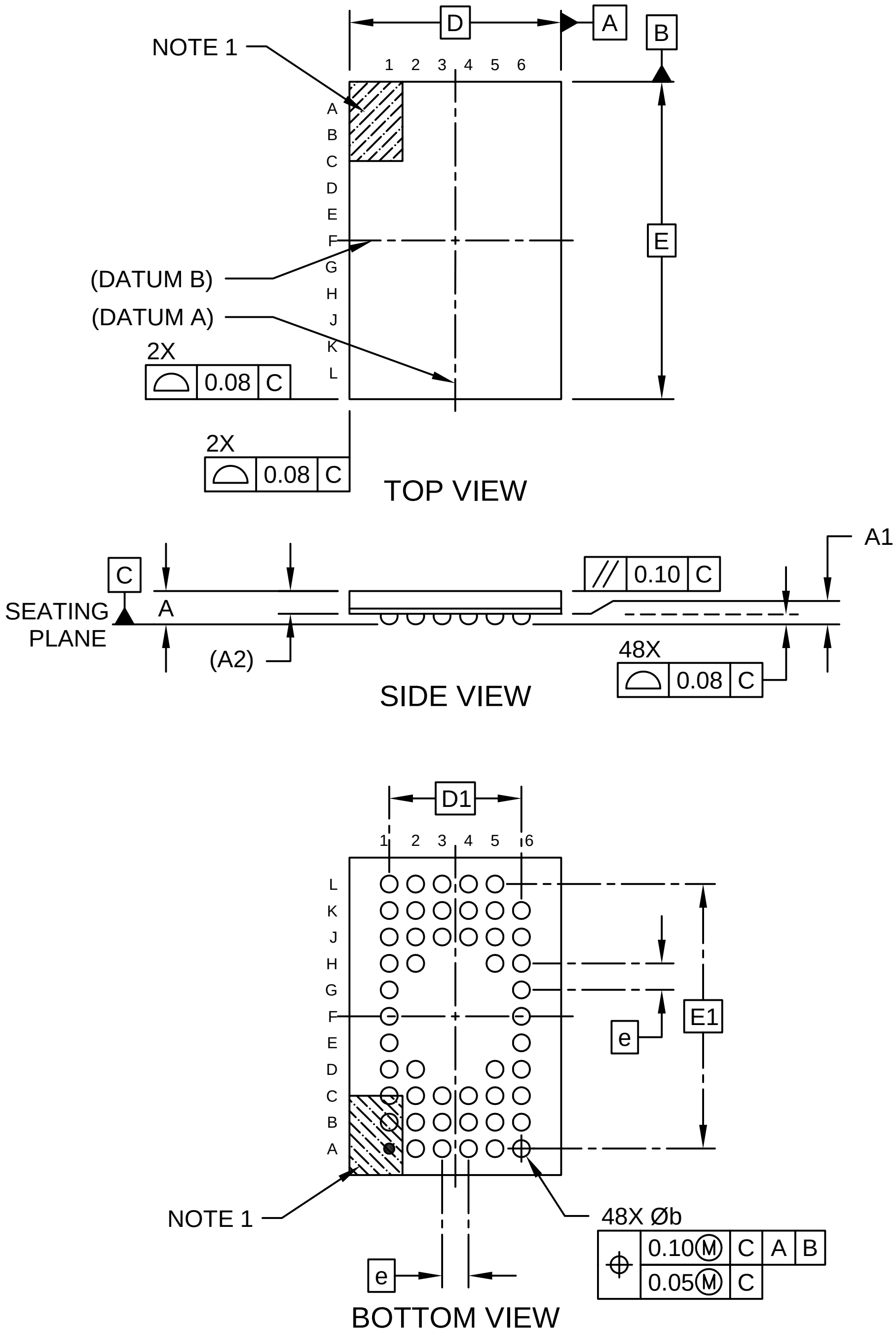
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2165-CM Rev B

48-Ball Very, Very Thin Fine Pitch Ball Grid Array (CQ) - 4x6 mm Body [WFBGA] SST LEGACY MAQE and MAQF

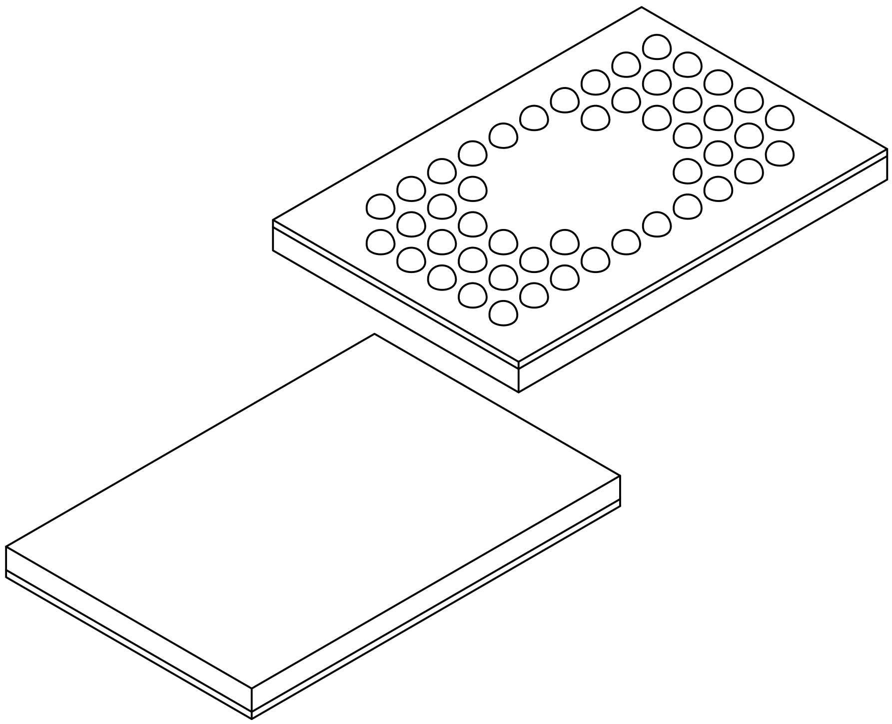
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-165-CQ Rev B Sheet 1 of 2

48-Ball Very, Very Thin Fine Pitch Ball Grid Array (CQ) - 4x6 mm Body [WFBGA] SST LEGACY MAQE and MAQF

Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		48	
Pitch	e		0.50 BSC	
Overall Height	A	0.53	0.63	0.73
Standoff	A1	0.14	0.20	0.26
Package Thickness	A2		0.43 REF	
Overall Length	D		4.00 BSC	
Overall Terminal Spacing	D1		2.50 BSC	
Overall Width	E		6.00 BSC	
Overall Terminal Spacing	E1		5.00 BSC	
Terminal Width	b	0.27	0.32	0.37

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

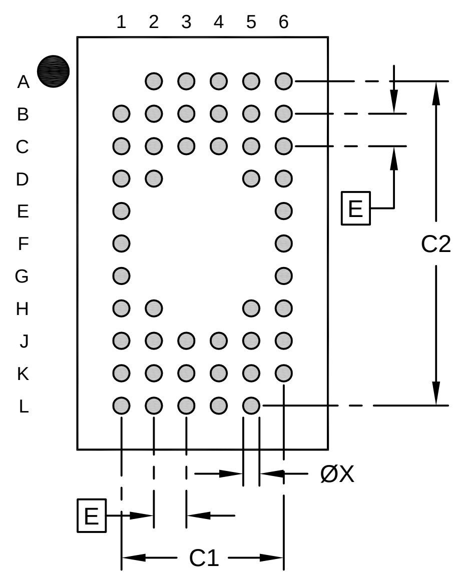
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-165-CQ Rev B Sheet 2 of 2

Land Pattern

48-Ball Very, Very Thin Fine Pitch Ball Grid Array (CQ) - 4x6 mm Body [WFBGA] SST LEGACY MAQE and MAQF

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Contact Pad Spacing	C1		2.50	
Contact Pad Spacing	C2		5.00	
Contact Pad Diameter (X48)	X			0.25

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2165-CQ Rev B